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Measurement Without Validity: The Compounding Reliability Problem in Agentic AI Evaluation

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Abstract

Agentic AI evaluation pipelines produce benchmark scores that justify deployment decisions, safety certifications, and regulatory compliance claims. No formal framework has characterized how validity degrades across these stages. We present a knowledge-based three-layer compounding validity model, $V_{\text{total}} \leq V_1 \times V_2 \times V_3$, that captures multiplicative degradation across task generation (V_1), human-simulator calibration (V_2), and automated judgment (V_3). Applying it to five widely-used agentic benchmark configurations, we find V_1 and V_2 independently measurable and both degraded, but V_3 not currently quantifiable for any: three of the four benchmarks score with a single deterministic rule, so no second rater exists and reliability is undefined, and the fourth's partial LLM grader has no published grader-vs-human measurement.

A structured survey of 55 published agentic evaluation papers finds that 26 of them report no chance-corrected inter-rater statistic of any kind, a determination four independent raters recover at Krippendorff's $\alpha = 0.834$. On the broader judgment of whether a reported metric fits its rating design, approximately 80% of this purposive sample use structurally mismatched, incomplete, or absent inter-rater reliability (IRR) metrics (four-rater range 60.7%–84.4%). Both are consistent with systematic V_3 collapse. We further document V_1 failures (task validity flaws in 7 of 10 benchmarks) and V_2 miscalibration (up to 9 percentage points inter-simulator variance, with systematic demographic disparities for non-Standard American English speakers).

We derive eight prescriptions grounded in psychometric science and

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domain-stratified reliability thresholds (intraclass correlation $\geq$0.70; $\alpha \geq$ 0.67/0.70/0.80 by consequence level) that practitioners can apply immediately.

Keywords: inter-rater reliability, agentic AI evaluation, construct validity, LLM-as-a-Judge, knowledge-based evaluation framework, decision support, benchmark quality

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1. Introduction

Deployment decisions, safety certifications, and regulatory compliance claims for agentic AI systems are all downstream of a number that appears deceptively clean: a benchmark score. That score is the product of a pipeline (task generation, world simulation, rater judgment), and each stage of that pipeline is a potential source of systematic, non-random measurement error.

The measurement problem in agentic evaluation is not simply that individual components are imperfect. It is that the imperfections compound. A pipeline that loses 30% of valid signal at task generation, 20% at simulation, and 25% at the judgment layer does not produce evaluation that is 75% valid. Under the independence bound, it produces evaluation that retains less than 42% of valid signal against the intended construct. This bound is itself optimistic for same-provider pipelines, where shared systematic biases not captured in the per-layer estimates cause them to overstate true validity (Appendix B, §B.4).

This paper makes three contributions. First, we document empirically that agentic evaluation fails three distinct psychometric validity tests simultaneously, and that the empirical evidence for each failure is published, peer-reviewed, and specific. Second, we introduce a heuristic three-layer compounding validity model that characterizes how failures multiply rather than add, a relationship the field has not yet quantified, and which we present as a conceptual bound rather than a proved theorem. Third, we derive eight prescriptions from the psychometric literature that practitioners and benchmark authors can apply immediately.

We are not arguing that agentic evaluation is worthless. We are arguing that it is less valid than current practices acknowledge, that the mechanisms of invalidity are known, and that the standards for reporting and interpreting evaluation results should be substantially higher.

2. Background: The Psychometric Standard

This section introduces the two psychometric constructs that underpin the analysis: construct validity, which defines what a measurement system is supposed to capture, and inter-rater reliability, which quantifies the consistency of the scoring process. Both are necessary conditions for a valid benchmark score.

2.1. Construct Validity as the Organizing Concept

The foundational criterion for any measurement system is construct validity: whether the measurement instrument actually captures the phenomenon it claims to measure [1]. The measurement science tradition identifies three evaluable validity types that together constitute evidence for or against construct validity:

- **Content validity**: does the evaluation cover the full scope of the construct, or only an easily measurable subset? Does a benchmark described as "measuring agentic tool use" sample the full range of tool-use behaviors, or does it measure only the subset most amenable to automated scoring (e.g., API call success rate)?
- **Criterion validity** (concurrent and predictive): does the evaluation score correlate with an external criterion of the target construct? Predictive criterion validity specifically asks whether the score predicts real-world deployment performance.
- **Construct validity evidence** (convergent and discriminant): does the evaluation correlate with conceptually related measures (convergent evidence) and fail to correlate with conceptually unrelated ones (discriminant evidence)?

Despite its centrality to measurement science, construct validity has received limited attention in AI evaluation, even as benchmark creation involves numerous high-stakes design decisions including target user selection, construct-to-task operationalization, and scoring metric choice [2]. A systematic review of 84 academic and industry papers found that technical performance was represented in 83% of evaluated works, while human-centered evaluation appeared in only 30%, and both were combined in only 15% [3]. Benchmark success and deployment value remain systematically disconnected.

Content validity failures are measurable and consequential. A systematic analysis of 194,955 benchmark questions mapped against the EU AI Act's taxonomy of model capabilities found that current benchmarks devote 61.6% of regulatory-relevant coverage to hallucination tendency and 31.2% to performance reliability, while capabilities central to loss-of-control scenarios (including evading human oversight, self-replication, and autonomous AI development) receive zero coverage across the entire public benchmark corpus [4].

2.2. Inter-Rater Reliability: Metrics and Their Conditions of Use

Inter-rater reliability (IRR) quantifies the degree to which raters (human annotators, LLM judges, or Agent-as-a-Judge systems) agree on the same outcome. The appropriate IRR metric is determined by the structure of the rating design, not by convention or familiarity.

Cohen's κ [5] is appropriate when exactly two fixed raters score every item using the same two-rater configuration across all items. It corrects for chance agreement using per-rater marginal distributions.

Fleiss' κ [6] generalizes to three or more raters, with the critical property that rater identity may vary per item provided the number of ratings per item is constant. It uses the pooled distribution across all raters for its chance-agreement correction. This is the correct metric for fixed-size judge panels where individual judge identity may rotate.

Krippendorff's α [7] is the most general metric: it handles any number of raters with varying identity, supports missing data natively, and accommodates nominal, ordinal, interval, and continuous measurement scales through interchangeable distance functions. For ordinal rubrics, it uses ranked distance rather than binary mismatch, correctly weighting the severity of disagreement.

Intraclass Correlation Coefficient (ICC) extends IRR concepts to continuous outcomes. The two-way mixed-effects absolute agreement form, $\mathrm{ICC}(A, 1)$, measures the degree to which scores from one rater (or measurement occasion) agree in absolute value with scores from another. Unlike κ-family metrics, $\mathrm{ICC}(A, 1)$ is appropriate when outcomes are measured on a continuous or interval scale and when absolute agreement (not merely consistency of ranking) is the validity target. In the context of this paper, it is used to quantify simulation calibration: the agreement between agent success rates under simulated versus real user conditions (V_2, Section 5).

Table 1 summarizes metric selection by scenario.

Table 1: IRR metric selection by agentic evaluation scenario.

Scenario	Recommended metric	Reason
3+ LLM judges, binary, complete matrix	Fleiss' κ	Nominal data, complete, multi-rater
3+ LLM judges, 1–5 rubric, complete matrix	Krippendorff's α (ordinal)	Ordinal distance metric needed
3+ LLM judges, continuous scores, complete matrix	Krippendorff's α (interval)	Interval distance required
2 fixed raters, categorical labels, every item	Cohen's κ	The two-rater nominal case, and the only one Cohen's κ fits
2 fixed raters, ordinal labels, every item	weighted Cohen's κ or Krippendorff's α (ordinal)	Two raters, but an ordinal scale needs a distance function; unweighted Cohen's κ here is the Mode 2 error of §6.2
2 fixed raters, continuous scores, every item	ICC$(A, 1)$	Two-rater absolute agreement on a continuous scale
Rotating judge panel, constant number of judges per item	Fleiss' κ (nominal) or Krippendorff's α by scale	Identity may rotate; what matters is that the count per item is constant
Ratings missing, or the number of raters per item itself varies	Krippendorff's α (general)	The only design requiring the general form

2.3. Reliability Thresholds and Their Origins

The Landis–Koch scale [8] is the most frequently cited threshold system in AI evaluation: $\kappa > 0.21$ as "fair," $\kappa > 0.41$ as "moderate," $\kappa > 0.61$ as "substantial." This scale was developed for behavioral science contexts with human raters and has no special authority in AI evaluation. Its widespread use as an acceptability benchmark for LLM judge agreement is a category error.

For agentic evaluation, domain-calibrated thresholds grounded in construct validity theory are more appropriate. We propose the following tiered thresholds, derived in Section 8:

- **Exploratory capability evaluation**: $\alpha \geq 0.67$ [7, pp. 241–243]
- **Safety, bias, fairness, and risk scoring**: $\alpha \geq 0.70$ (*proposed; see Prescription 5*)
- **Scoring that gates deployment, compliance, or regulatory reporting**: $\alpha \geq 0.80$ (*proposed; see Prescription 5*)

3. Related Work

3.1. Evaluation Documentation and Knowledge Frameworks

Structured knowledge frameworks for AI evaluation documentation have emerged as a response to inconsistent reporting practices. Mitchell et al. [9] introduced Model Cards, encoding expert knowledge about model behavior, intended use, and limitations into a standardized artifact that practitioners can apply without deep ML expertise. Dhar et al. [10] extended this to EvalCards, a structured template for evaluation-methodology disclosure organized around reproducibility, accessibility, and accountability. It does not yet require an inter-rater reliability field, which is precisely the addition Prescription 8 proposes. These frameworks follow the knowledge-based systems tradition of encoding expert domain knowledge into structured decision procedures that non-specialists can apply consistently [11, 12]: just as early medical expert systems (MYCIN, INTERNIST) encoded diagnostic knowledge into rule-based structures, documentation frameworks encode evaluation methodology expertise into required reporting fields. The compounding validity model ($V_{\text{total}} \leq V_1 \times V_2 \times V_3$) and the IRR metric selection decision tree proposed in this paper complement these documentation standards by providing the underlying validity model that determines *what* to report before documentation is written.

3.2. Validity Theory Applied to AI Evaluation

The foundational measurement science concept this paper applies is construct validity [1]: whether a measurement instrument captures the phenomenon it claims to measure. Despite its centrality, construct validity has received limited attention in AI evaluation practice [2]. Jacobs and Wallach [13] demonstrated that fairness metrics routinely measure constructs different from their stated targets, a specific instance of the construct validity failure this paper characterizes at the pipeline level. Meimandi et al. [3] provided empirical evidence that agentic AI evaluation systematically fails to support the productivity claims it is used to justify. That result is about which dimensions are measured rather than about task-distribution fidelity, so it bears on the content-validity argument of Section 2.1 rather than on V_1 directly. The three-layer compounding model operationalizes construct validity theory for multi-stage automated pipelines, providing a tractable computational framework for what has previously been treated as a qualitative concern.

3.3. Decision Support for Reliability Metric Selection

The IRR metric selection decision tree (Figure 1) follows the expert system tradition of encoding domain expertise into formal decision procedures that practitioners can apply without deep methodological background [7]. James [14] provided a systematic guide to inter-annotator agreement metric selection in NLP annotation, arguing against one-size-fits-all use of Cohen's κ, a position directly aligned with the prescriptions in Section 8. The decision tree proposed here extends this guidance to agentic AI evaluation pipelines, encoding the structural conditions (rater count, identity stability, measurement scale) that determine metric validity into a formal procedure. This knowledge representation is the framework's primary deployable artifact: it transforms implicit expert knowledge about IRR metric selection into an explicit, auditable decision structure.

4. Layer 1: Task Generation Validity

In traditional static benchmarks, tasks and ground truth are fixed. Reliability can be validated once against a stable reference. In agentic evaluation, tasks are often dynamically generated by language models, embedding systematic biases from the generator into the evaluation distribution before a single agent response is scored.

The construct validity problem at this layer is that dynamically generated tasks may not sample the intended construct uniformly. An LLM task generator that overrepresents certain task types, difficulty levels, or surface forms creates a distribution that raters can agree on perfectly, while measuring something systematically different from the intended capability.

The empirical evidence is specific. A preprint study (under review at time of submission) evaluating 10 popular agentic benchmarks found task validity flaws in 7 of them and outcome validity flaws in 7 of them, with all 10 showing benchmark reporting gaps [15]. Documented failures include task validity failures where do-nothing agents passed 38% of airline booking tasks, and outcome validity failures where LLM judges made arithmetic errors. Every benchmark showed at least one validity failure category; none provided sufficient evidence to support unqualified capability claims.

Task generation validity must be assessed before any IRR analysis is meaningful. A rating matrix derived from systematically biased tasks cannot be rescued by metric sophistication at the judgment layer.

5. Layer 2: Simulation Calibration

Many agentic benchmarks evaluate agents in interaction with simulated users or simulated world environments rather than real humans and real environments. This design choice is operationally sensible (real human interaction at evaluation scale is expensive and slow), but it introduces a second, distinct validity problem: whether the simulated environment faithfully proxies the real deployment context.

5.1. Empirical Evidence of Simulation Miscalibration

The most direct empirical evidence comes from Seshadri et al. [16], which tested LLM user simulation against real human users on τ-Bench retail tasks across participants in the United States, India, Kenya, and Nigeria. The findings establish simulation miscalibration as a measured fact rather than a theoretical concern:

- Agent success rates varied up to 9 percentage points across different LLM user simulators, demonstrating that simulation results are not robust to simulator choice.

- Evaluations using simulated users exhibited systematic directional miscalibration: underestimating agent performance on challenging tasks while overestimating it on moderately difficult ones.
- Simulated users introduced conversational artifacts absent from real interactions: elevated question-asking, heightened politeness markers, and artificial turn structures.

5.2. The Demographic Asymmetry Problem

The calibration failures documented by Seshadri et al. [16] are not uniformly distributed across user populations. African American Vernacular English (AAVE) speakers experienced consistently worse success rates and calibration errors than Standard American English (SAE) speakers, with disparities compounding with age. Indian English speakers showed similar patterns.

This is simultaneously a measurement validity failure and a fairness failure. An evaluation pipeline calibrated on SAE simulated users is not a valid measurement of agentic capability for AAVE-speaking or non-SAE users. As shown in Section 6, the same demographic asymmetry is compounded at the judgment layer by LLM judges that exhibit systematic miscalibration across dialect groups independently of simulation layer failures.

5.3. Why Agreement on a Distorted Signal Does Not Establish Measurement Validity

The key inferential step at this layer is often missed: high rater agreement on simulated interactions does not demonstrate evaluation reliability. It demonstrates that all raters agree on a distorted input.

A scoring panel that perfectly agrees on agent scores produced by an AAVE-miscalibrated simulator has achieved high IRR in the technical sense. It has also produced a high-reliability measurement of the wrong thing. Agreement on a distorted signal is agreement on distortion.

This distinction matters for how κ is used as a V_3 proxy throughout this paper (Section 7.2). Where an external human criterion is available, κ between a grader and that criterion measures the grader's *consistency* with an independently established reference, not correctness in an absolute sense; this is the standard, imperfect way the field operationalizes grader validity, and it is what the V_3 estimates in Table 3 are intended to capture. This is a different measurement target from the raters-agreeing- on-a-distorted-input problem

above, where no external criterion is involved and agreement is diagnostic of nothing beyond the raters' shared exposure to the same distortion.

6. Layer 3: Metric Misspecification

The third layer of validity failure occurs at judgment: even when a human or LLM judge scores agent outputs, the inter-rater reliability metric used to validate that scoring is structurally mismatched to the pipeline design in the large majority of published evaluations.

6.1. Why Cohen's κ Is Structurally Wrong for Most Agentic Evaluation Designs

Cohen's κ requires exactly two raters scoring every item, with the same two-rater pair maintained across all items. Almost no agentic evaluation design satisfies this constraint:

- LLM-as-a-Judge pipelines typically use a rotating pool of judge models that vary across evaluation runs or item subsets.
- Agent-as-a-Judge frameworks [17] replace the rater with a multi-component agentic pipeline whose judgment is assembled from several interacting modules, so "rater identity" is not a fixed quantity that a two-rater chance correction can condition on.
- Human annotation at scale uses crowdsourcing pools where rater identity varies per item.
- Dynamic benchmark generation produces item batches scored by different annotator subsets.

Applying Cohen's κ to any of these designs produces a mathematically invalid result, because its chance-agreement correction assumes fixed rater identity. Using it anyway (common in practice) is a silent validity failure that produces an uncorrectable reliability estimate without warning.

6.2. The Literature Scan: IRR in Published Agentic Evaluation Papers

Methodology. We conducted a structured survey of 55 papers published or posted between 2022 and 2026, selected via a stratified purposive approach across nine topic categories: major agentic benchmarks; automated grader validity studies; benchmarks with correct IRR use; LLM-as-a-Judge studies; benchmarks with structural metric mismatch; long-horizon evaluation; safety and RLHF annotation; recent 2025–2026 evaluation papers; and safety-critical IRR failure cases. Within each category, papers were selected for citation prominence, venue tier (ACL, EMNLP, NeurIPS, ICLR), and direct relevance to evaluation methodology; arXiv preprints in cs.AI and cs.CL supplemented peer-reviewed coverage where thin. This is a purposive stratified survey, not a systematic review: the sample is designed to cover the main failure modes rather than to exhaust a keyword-defined corpus. The complete per-paper coding table is given in Appendix A (Table A.2).

Coding was applied against four pre-specified dimensions: (1) the IRR metric reported or absent, (2) the rater design (number of raters, fixed or rotating identity), (3) the measurement scale (binary, nominal, ordinal, or continuous), and (4) whether the metric's structural assumptions were satisfied by the pipeline design. The full coding criteria are documented in Appendix A.

Reliability validation. Coding was performed by the primary author. To assess coding reliability, all 55 papers in the corpus were coded independently by three LLM raters from distinct model families: NVIDIA Nemotron-Ultra-550B, Google Gemma-4-31B, and Alibaba Qwen3-80B-A3B [18]. LLM raters were chosen deliberately rather than as a substitute for an unavailable human coder: LLM-as-a-Judge is the prevailing rating pattern in commercial and academic benchmarks, and it is precisely the practice whose reliability disclosure this paper argues is inadequate. The check therefore applies the paper's own prescriptions to that practice: a cross-family panel (Prescription 4), the structurally correct metric for a four-rater, complete-matrix nominal design (Fleiss' κ, Rule R3 of Table A.3), reported alongside Krippendorff's α (nominal), and full reporting of the result including pairwise breakdowns (Prescription 8). We lead with α for comparability with the thresholds of Section 2.3 and report the R3-prescribed κ beside it; on a complete matrix of this shape the two coincide to within 0.001, which is itself a check that neither was mis-specified. Each LLM rater received the coding instrument together with the full extracted text of the paper under coding, with no access to the primary-rater codes. The stimulus is the entire document rather than

a curated excerpt, so that locating the relevant evidence is part of the coding task rather than a step already performed for the rater. Each rater coded all 55 papers in five independent runs, and the author's codes cover all 55 papers, so the design is a genuinely complete four-rater matrix.

Because each rater was asked to quote the sentence its code rested on, the returned evidence can itself be checked against the stimulus. A released probe (`validate.py`) looks for each quoted run of words in the document the rater was coding. It locates 89% of the quotes for the metric-identification dimension, 85% for rater design and 88% for measurement scale; a further 8–10% cannot be found in the source at all. Those are codes delivered with evidence that does not exist in the document, and they are reported here rather than filtered out, because a rater that supplies a confident code attached to an unlocatable quotation is a sharper instance of this paper's argument than any of the coefficients below.

Reliability differs sharply by what is being coded. On the metric-identification dimension (Q1: which coefficient, if any, the authors computed on their own annotations), four-way agreement is Krippendorff's $\alpha = 0.834$ (median across the five runs; range 0.818–0.846), with Fleiss' $\kappa = 0.818$–0.846, at or above the $\alpha \geq 0.80$ level this paper proposes for deployment-gating use. On the three interpretive dimensions it is substantially lower: $\alpha = 0.371$ for rater design (Q2), $\alpha = 0.525$ for measurement scale (Q3), and $\alpha = 0.536$ for the structural-validity judgment itself (Q4), the dimension that assigns each paper a failure mode. Collapsing the failure-mode code to the binary distinction that generates this paper's headline figure, correct use versus misuse, gives $\alpha = 0.549$, with the four raters placing the misuse rate between 60.7% and 84.4% around the author's 80.0%. Within-rater stability across the five runs was 100% (Gemma), 98% (Qwen), and 95% (Nemotron) on the metric-identification dimension, so the between-rater figures are not an artifact of sampling noise. That stability is markedly higher than Haldar and Hockenmaier [19] report for LLM judges scoring open-ended generation, where agreement of a judge with itself across repeated runs is low enough that they describe the resulting ratings as almost arbitrary in the worst case. The contrast supports the division this section documents: a judge asked which coefficient a paper computed is answering a question whose answer is printed in the paper, whereas a judge scoring generation quality is supplying the standard as well as applying it. Wei et al. [20] reach the same separation from a different direction: they model a judge's run-to-run variation as a flip probability, report it as a self-consistent rate of 0.91–0.98 across temperature

settings and both of their datasets, and then remove it from their agreement estimate rather than letting it inflate one. The principle that within-rater instability and between-rater disagreement are different quantities is therefore not particular to this study.

This division is informative rather than incidental. Determinate questions, whose answer is printed somewhere in the paper, are coded reliably; judgments requiring a coder to decide whether a rating design satisfies a metric's structural assumptions are not. The pairwise breakdown shows the same pattern: on metric identification (Q1) the author agrees with Gemma at $\alpha = 0.924$, Nemotron at 0.873, and Qwen at 0.750, whereas on the structural-validity judgment (Q4) the corresponding values fall to 0.731, 0.604, and 0.412. We therefore report the 80% misuse rate with its reproducibility range attached rather than as a point estimate, and Section 10.2 discusses what the residual disagreement reveals. The pairwise matrices for the other two dimensions are released with the per-dimension reports rather than reproduced here; on rater design in particular the author and Qwen agree at $\alpha = 0.029$, which is the weakest cell anywhere in the study.

Finding. The most securely established result is the simplest one: 26 of the 55 papers, 47%, report no chance-corrected inter-rater statistic of any kind. That is a determination about which coefficient appears in a paper, the dimension four independent raters recover at $\alpha = 0.834$, and it does not depend on anyone agreeing about what counts as misuse. Every claim this paper makes about the judgment layer follows from it alone.

The broader characterization is that 11 papers (20%) apply a structurally correct IRR metric with a stated threshold and rationale, and the remaining 44 (80%) fall into one of four failure modes. These follow the structural-validity code of the coding instrument, with its INC code separated into percent-agreement-only and correct-coefficient-without-threshold cases (Appendix A, Table A.2). This 80% figure carries the reliability caveat established above, and two further qualifications belong with it. First, five of the 55 conduct no rating or annotation study of their own, so there is nothing for which they could have reported reliability: a metric selection guide, a benchmark checklist, a survey, and two theoretical papers. Restricting the rate to the 50 papers that do rate something gives 78.0% (39 of 50), which is the more conservative reading and the one we would defend. Second, the categories were designed to span failure modes rather than to sample the field, which inflates the rate further (Section 10.2):

Mode 1: No IRR Metric Reported (26 papers, 47%).. Within this largest group, two sub-patterns recur. The first is automated grading with no human reliability measurement: major benchmark papers (τ-bench [21], OSWorld [22], SWE-bench [23], GAIA [24], and WebArena [25]) use deterministic automated graders or unvalidated LLM judges with no IRR measurement against human annotation; τ^2-Bench [26], the quality-fix follow-up to τ-bench, takes the same approach, tighter automated grading rather than human IRR validation. Gurram [27] (preprint, under review at time of submission) directly measures the gap such graders leave, and its full set of coefficients is more informative than any one of them: substring-based automated evaluation reaches $\kappa = 0.049$ against each of two human annotators, essentially chance level; a three-LLM ensemble reaches $\kappa = 0.432$; a single GPT-4o-mini judge does better still at $\kappa = 0.567$; and the two human annotators agree with each other at $\kappa = 0.835$. Automated grading therefore sits well below the human–human ceiling even at its best, and the ensemble is not the strongest automated option, so the gap is not closed simply by adding judges. The Verified re-release of WebArena (WebArena Verified, El Hattami et al. 28: Cohen's $\kappa = 0.83$ on its task-audit labels; Section 6.5) shows that rigorous human annotation produces fundamentally different reliability evidence, even where the labeled construct is benchmark quality rather than agent outcome. The second sub-pattern is evaluation that reports a statistic which is not an IRR metric, or no rater study at all: Pearson correlation for ordinal safety ratings from 112 demographically diverse annotators [29]; ELO and Bradley–Terry ratings substituting for inter-annotator reliability in crowdsourced pairwise preference [30]; and RLHF annotation pipelines, red-teaming reports, surveys, and position papers in which the absence of reliability reporting is treated as unremarkable. In every case the absence of any reliability report is a complete V_3 validity gap.

This sub-pattern is not an artifact of the sampling frame, and it reaches work written specifically to improve judge reliability. Wei et al. [20], which is outside this scan's agentic scope and so is not one of the 26 papers counted above, sets out to give LLM-as-a-Judge "more theoretically interpretable evaluation metrics" and to mitigate judge internal inconsistency. It reports judge agreement with human evaluators as raw accuracy, mostly below 0.7, and introduces no chance-corrected coefficient at any point. A reader cannot tell from an accuracy of 0.68 whether the judge is tracking the human standard or tracking the marginal distribution of the labels, which is the distinction a chance correction exists to make. When a paper devoted to measuring

judge reliability still omits that correction, the omission is better read as a convention of the field than as an oversight by individual authors.

Mode 2: Structural Metric Mismatch (11 papers, 20%).. Papers that do report a chance-corrected metric most often apply one whose structural assumptions the rating design does not satisfy. Five apply an unweighted κ-family statistic to an ordinal or ternary scale without a distance correction (Fan et al. 31 AgentProcessBench: unweighted Cohen's $\kappa = 0.767$ on a ternary $\{-1, 0, +1\}$ scale, alongside the commonly-cited 89.1% raw agreement figure; Ganguli et al. 32: Fleiss' $\kappa = 0.32$ on a five-point Likert scale). Four apply an inter-rater metric (Cohen's κ, Fleiss' κ, Krippendorff's α, or ICC) to repeated stochastic runs of a *single* model, a test–retest self-consistency design in which no second rater exists and the chance-agreement correction is undefined (Table A.2, entries 6, 8, 46, and 47). In Han et al. [33] the mismatch is more subtle: Cohen's κ is applied as a series of pairwise comparisons (each of 54 LLMs vs. human ratings), each individually a valid two-rater design, but the results are then aggregated and interpreted as panel reliability across 54 configurations with varying rater compositions, a use case requiring Fleiss' κ or Krippendorff's α, not repeated two-rater κ. The same averaging pattern recurs in a hate-speech reliability study that computes Cohen's κ for all annotator pairs and reports the average across three annotators drawn from rotating crowdsourced pools (Table A.2, entry 18); because rater identity varies across items, the design calls for Krippendorff's α or Fleiss' κ rather than an averaged two-rater κ. That paper explicitly argues that Cohen's κ is improper for subjective labeling tasks and then applies it regardless, which illustrates how weakly metric choice is currently constrained even by authors alert to the problem.

What makes both of these a mismatch is that the averaged pairs rate the *same* items. Averaging pairwise κ over a shared item set conflates variance between rater pairs with agreement about items, which is precisely the case Fleiss' κ and Krippendorff's α exist to handle. Aggregating per-pair coefficients computed on *disjoint* item sets is a different operation and is not a mismatch: WebArena Verified (Section 6.5) does the latter, reporting a task-weighted overall κ across four site-specific fixed pairs alongside the per-site values from which it is formed. Each of its site-level values is in turn a macro-average across annotation categories on the same items, which is a different averaging axis and not the one at issue here: the categories are distinct constructs each rated by the same fixed pair, not competing raters of

one construct.

Mode 3: Percent Agreement Only, No Chance Correction (5 papers, 9%).. The remaining papers report raw agreement with no correction for chance: 40 contractors ranking model outputs [34, 72–77%]; a 95.9% inter-annotator agreement figure for binary web-agent outcome labels (Table A.2, entry 10); an 85% two-reviewer screening agreement [3]; raw LLM–human accuracy for dialectal toxicity judgments [35]; and an approximately 85% agreement rate between an LLM judge and human labels for web-agent success (entry 39). Raw agreement is uninterpretable without a chance baseline: for a binary label with skewed marginals, high percent agreement is compatible with near-zero κ.

Mode 4: Correct Coefficient, Incomplete Reporting (2 papers, 4%).. Two papers select a structurally appropriate coefficient and then report it without the threshold or rationale that would make the value interpretable. A large-scale audit of annotation reporting computes a macro-averaged Krippendorff's $\alpha = 0.585$ for its own annotation study and calls that "decent" agreement, a characterization corresponding to no stated convention: Krippendorff's own guidance treats $\alpha \geq 0.800$ as the threshold for reliable conclusions and 0.667 as the floor for tentative ones, and 0.585 clears neither (Table A.2, entry 19). A study of length bias in preference evaluation reports Fleiss' $\kappa = 0.66$–0.74 across five raters per segment, labels that range "moderate to substantial" without citing a banding scheme, and generalizes a reliability claim from an inter-rater analysis covering only one of its evaluation settings (entry 43). This is the narrowest mode in the scan and the most direct instance of the paper's central claim: the coefficient is the correct one and the arithmetic is sound, yet no interpretive commitment is attached to the number, so the reported value constrains no conclusion.

Table 2 summarizes the distribution.

Structural contrast. The 11 papers using IRR correctly share a distinguishing behavior: they state the rationale for the metric chosen, and several report more than one coefficient. Papers coded as mismatched typically report a single coefficient with no structural justification. The mismatch does not read as a deliberate simplification; it reflects unawareness of the structural conditions that govern metric validity.

Table 2: IRR failure modes across 55 individually coded papers, by the structural-validity code of the coding instrument (Appendix A). Only 11 (20%) use structurally correct metrics with explicit rationale.

Failure mode	Papers	%	Representative example
No IRR metric reported	26	47	τ-bench, OSWorld, SWE-bench (automated grading); Chatbot Arena (ELO); RLHF and red-teaming reports
Structural metric mismatch	11	20	Cohen's κ for 54-LLM panel [33]; averaged pairwise Cohen's κ over rotating annotators (entry 18); Fleiss' κ on a Likert scale [32]; α on repeated runs of one model
Percent agreement only	5	9	InstructGPT (72–77%); Emergence WebVoyager (95.9%)
Correct coefficient, incomplete reporting	2	4	Krippendorff's $\alpha = 0.585$ called "decent" with no stated convention (entry 19); Fleiss' $\kappa = 0.66$–0.74 banded without citation (entry 43)
Correct metric	**11**	**20**	WebArena Verified ($\kappa = 0.83$, on benchmark-audit labels rather than agent outcomes); MCP-AgentBench (Fleiss' $\kappa = 0.671$)

6.3. Demographic Asymmetry in LLM Judge Calibration

The metric mismatch problem documented above is compounded by a systematic calibration asymmetry in LLM judges across linguistic varieties. In a study evaluating LLMs as toxicity judges across 60 language varieties spanning 10 language clusters, LLM–human agreement was the weakest dimension of consistency (weaker than cross-model or cross-dialect consistency), with systematic gaps across dialectal groups [35]. Judge miscalibration for non-standard dialect speakers persists in direct LLM judgment, independently of the simulation layer failures documented in Section 5.2.

The implication for the compounding model is asymmetric V_3: benchmarks using LLM judges without dialect-stratified calibration validation will produce V_3 values that overstate validity for SAE-speaker user populations and understate it for non-SAE populations. This is an IRR validity failure that structurally correct metric selection cannot by itself correct, because the miscalibration occurs within the judge's learned behavior, not in the choice of metric.

6.4. Ordinal Rubrics and the Kappa Penalty

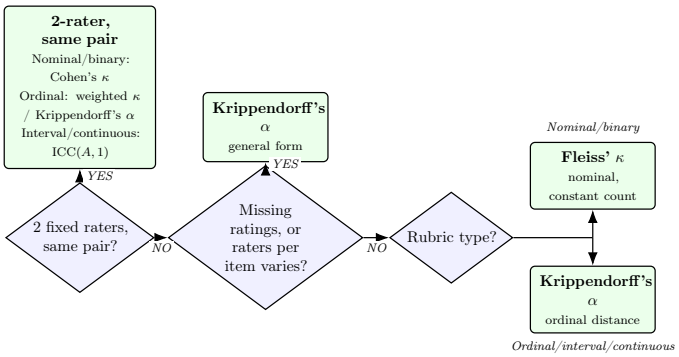
Most agentic evaluation rubrics use ordinal scoring (e.g., 1–5 on helpfulness, safety, coherence). Fleiss' κ treats all disagreements as categorically equal: a rater scoring 1 when others score 5 is penalized identically to a rater scoring 4 when others score 5. This is wrong for ordinal data.

The consequence is bidirectional distortion: Cohen's κ and Fleiss' κ artificially inflate apparent disagreement when raters are closely aligned on an ordinal scale (one point apart) and can underestimate disagreement when raters are at opposite ends of the scale. For safety evaluation, this matters: a safety rubric where judges disagree by 1 point appears as unreliable as one where they disagree by 4 points.

6.5. Metric Selection Decision Tree

Figure 1 provides a decision tree for selecting the structurally correct IRR metric based on rater design, data completeness, and measurement scale.

Figure 1 is a knowledge-based decision support system in the expert systems tradition [11, 12, 14]: it encodes domain expert knowledge about IRR metric structural assumptions into a formal decision procedure that practitioners can apply without background in psychometrics. The terminal nodes are executable recommendations corresponding to the six production rules given as Table A.3 in Appendix A, §A.5; the two-rater terminal resolves into



Selection rules:

Cohen's κ
Exactly 2 fixed raters, same pair, every item, nominal/binary.

Weighted κ / Krippendorff's α
Exactly 2 fixed raters, same pair, every item, ordinal.

ICC$(A, 1)$
Exactly 2 fixed raters, same pair, every item, interval/continuous.

Fleiss' κ
$\geq$3 raters (fixed or rotating), constant count per item, nominal/binary.

Krippendorff's α (general)
Missing ratings, or the number of raters per item itself varies.

Krippendorff's α (ordinal/interval)
$\geq$3 raters (fixed or rotating), constant count per item, ordinal/interval/continuous.

Using Cohen's κ outside the 2-rater fixed-pair case produces an invalid reliability estimate without warning.

Figure 1: IRR metric selection. Left: decision tree based on rater design, matrix completeness, and rubric scale. Right: selection rules summary. A rotating panel with a constant number of raters per item (e.g. a fixed-size rotating judge pool) follows the same path as a fixed panel of the same size (Fleiss' κ or Krippendorff's α, §2.2); Krippendorff's α (general) is reserved for designs where that count is itself irregular or ratings are missing. Using Cohen's κ outside the two-rater fixed-pair case is a silent validity failure.

three rules by measurement scale (R1, R1†, R1‡). A practitioner who provides the rater count, identity stability, and scale type obtains a structurally correct metric recommendation without needing to understand the mathematical foundations of chance-agreement correction. This procedural knowledge representation is the framework's primary deployable artifact, complementing the governance prescriptions in Section 8.

Worked example 1 (LLM-as-judge, ordinal rubric). An evaluation uses three LLM judges (GPT, Claude, Gemini) scoring agent responses on a 1–5 helpfulness rubric, with all judges scoring every item. Applying the rules: rater_count = 3 (not R1); complete matrix, no missing ratings (not R2); scale is ordinal (not R3). Rule R4 applies: Krippendorff's α with ordinal distance function. Reporting Cohen's κ for this design (a common practice) violates R1's two-rater constraint and produces an invalid estimate.

Worked example 2 (human annotation, binary labels). WebArena Verified [28] assigns one primary annotator per site across the four self-hosted sites (Map is excluded from their reliability analysis because it is not self-hosted) and re-labels 100% of tasks with a paired blind verifier (site pairs A↔B, C↔D), so each task is scored by a fixed pair on a binary label. Applying the rules: rater_count = 2 (per pair), same pair for every item in that pair's sites, complete matrix (R1). Cohen's κ is the correct metric, and the reported site-level values (0.81–0.85; task-weighted overall $\kappa = 0.83$, 95% CI [0.81, 0.85]) meet the deployment-gating threshold ($\alpha \geq 0.80$, Prescription 5) for the construct actually being rated. The thresholds of Section 2.3 are stated for α and read here against a κ; for a binary label on a complete two-rater matrix the two coefficients are close enough for that comparison to be informative, but it is an approximation and not an identity, and we flag it rather than let the substitution pass silently. That construct is task-specification ambiguity and category-validity labeling, not agent-outcome success or failure; this example illustrates a structurally correct application of Cohen's κ (Prescription 2), but the resulting κ cannot be substituted for V_3 in Table 3, since V_3 requires reliability on agent-outcome judgment specifically (Section 7.2).

7. The Compounding Argument

The full mathematical treatment, including the same-provider estimator-bias discussion and a practical estimation protocol, is in Section 9 and Appendix B.

7.1. Heuristic Validity Bound

Let $V_1 \in [0, 1]$ denote the task generation validity: the degree to which the generated task distribution matches the ideal construct distribution, defined as one minus the total variation distance between them. Because the ideal construct distribution $D_{\mathcal{C}^*}$ is a theoretical object and not directly observable, V_1 cannot be computed from first principles; in practice it is estimated by comparing generated tasks against a curated expert reference set (Appendix B, §B.6). Let $V_2 \in [0, 1]$ denote the simulation calibration validity, defined as the $\mathrm{ICC}(A, 1)$ between simulated and real agent outcomes on a matched task set, estimated separately per user population group. Let $V_3 \in [0, 1]$ denote the judgment validity: the appropriate IRR metric value (Krippendorff's α or Fleiss' κ, per the pipeline's measurement scale) for the rating design. As an agreement coefficient, V_3 measures the grader's consistency with an independently established reference rather than correctness in an absolute sense (Section 5.3); it is a proxy for judgment validity, not a direct measurement of it.

The overall construct validity of the evaluation pipeline is bounded above by:

$$V_{\text{total}} \leq V_1 \times V_2 \times V_3 \tag{1}$$

The bound holds under the assumption of independence between layer failures. When the same provider family operates at all three layers, shared systematic biases not captured in the cross-pipeline estimates of V_1, V_2, and V_3 cause those estimates to overstate true validity for that specific configuration. The independence bound is therefore optimistic for same-provider pipelines; the practical V_{total} for such configurations may fall below it (Appendix B, §B.4).

Figure 2 illustrates the three-layer pipeline with its validity estimates and the compounding formula.

7.2. Applying the Model to Published Benchmarks

Table 3 applies the compounding model to five widely-used agentic benchmark configurations using values drawn directly from the cited literature.

V_1 **sources**: these are derived from the per-benchmark failure rates reported by Zhu et al. [15], using the rule $V_1 = 1 - r$, where r is the reported rate and the midpoint is taken where a range is given. Three of the four are scoped more narrowly than the benchmark name suggests, and we state the scope rather than let the benchmark label carry it: do-nothing agents

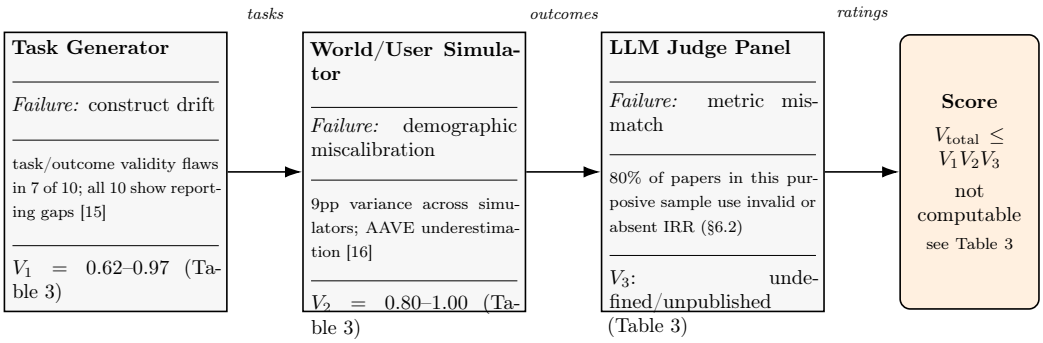


Figure 2: Three-layer agentic evaluation pipeline. Each layer introduces a distinct failure; failures compound multiplicatively under the independence bound ($V_{\text{total}} \leq V_1 \times V_2 \times V_3$). V_1 and V_2 are measured for the benchmarks in Table 3; V_3 is not, because three use a deterministic single-rule grader for which reliability is not a well-defined question, and the fourth has no published grader-vs-human measurement for its partial LLM-judgment component, so V_{total} cannot currently be computed for any of them. The bound itself is additionally optimistic for same-provider pipelines where shared systematic biases cause the external estimates to overstate true validity (Appendix B, §B.4).

passed 38% of τ-bench's airline subset ($V_1 = 0.62$); broken evaluation selectors caused 28% performance underestimation in OSWorld's Chrome task section ($V_1 = 0.72$); augmented test cases changed 41% of the SWE-bench Verified leaderboard, against 24.4% on Lite, and we take the Verified figure ($V_1 = 0.59$).

One of the four needs a stronger caveat. Zhu et al. [15] attribute WebArena's 1.4–5.2% misestimation to permissive string matching and to an LLM-as-a-Judge used without validating its accuracy, which under this paper's own three-layer division is a judgment-layer defect rather than a task-generation one. We retain $V_1 = 0.97$ for comparability with the other rows, but the figure should be read as an upper bound on WebArena's task-generation validity that happens to be constrained by a Layer-3 measurement, not as a measurement of Layer 1; and it sits in the same row whose V_3 we report as unpublished. More generally, Zhu et al. [15] classify the OSWorld and SWE-bench findings above as outcome-validity rather than task-validity issues, so only the τ-bench row is sourced from a finding their taxonomy and ours assign to the same layer. The compounding model's assumption that the three layers are independently estimable is therefore better supported for τ-bench than for the other three, and that is a limitation of the estimates in Table 3 rather than of the published measurements they draw on.

V_2 **sources**: OSWorld, SWE-bench, and WebArena use real computer or web environments without LLM user simulators; $V_2 = 1.00$ for these.

For τ-bench, Seshadri et al. [16] measured Expected Calibration Error ($\text{ECE}_{\text{Human-LLM}}$) between simulated and real user outcomes. We operationalize $V_2 \approx 1 - \text{ECE}/100$: 11.7% for Standard American English speakers ($V_2 = 0.88$) and 20.3% for AAVE speakers ($V_2 = 0.80$).

V_3 **sources**: V_3, as defined in this paper (Section 7.1), is an inter-rater reliability metric (κ or α), and inter-rater reliability presupposes at least two raters capable of disagreeing. τ-bench, OSWorld, and SWE-bench each score agent outputs with a single deterministic rule (database-state comparison, a custom execution script, and unit-test execution respectively): there is no second rater to disagree with. For these three benchmarks, V_3-as-reliability is not merely unmeasured; it is not a well-defined quantity under this model. A deterministic rule agrees with itself on every application, so it is trivially and perfectly self-consistent regardless of whether the rule correctly operationalizes the construct it is meant to check. WebArena (original) is mixed: its `exact_match` and `must_include` functions, which score the majority of its 812 tasks, share this same deterministic, no-second-rater property. Its `fuzzy_match` function, used for a subset of information-seeking tasks, is a genuine single-LLM judgment step, for which reliability is at least conceptually the right question, though no WebArena-specific grader-vs-human measurement has been published. Gurram [27]'s $\kappa = 0.049$ substring-matching estimate, measured on a different benchmark with a different grader implementation, would have to be borrowed by analogy rather than applied as a direct measurement, and even that analogy covers only the minority `fuzzy_match` portion of WebArena's tasks. We therefore do not compute V_3 or V_{total} for any benchmark in Table 3.

This gap is not merely a reporting omission. Deterministic, single-rule graders can be perfectly reliable while being invalid, and `must_include`-style substring checks illustrate why: a check for the substring "Paris" against the question "What is the capital of France?" scores the response "Paris is not the capital; Madrid is" as correct, every time, with zero disagreement possible, because the rule only tests for lexical presence rather than semantic correctness. No inter-rater reliability metric would ever surface this failure, because there is no second rater whose disagreement could flag it; the rule is reliable and wrong simultaneously. This is the deterministic-grader analogue of the argument in Section 5.3: agreement (or, here, trivial self-agreement) on a distorted signal is not evidence of validity.

Caveats on derived estimates. The V_1 and V_2 values above are derived

Table 3: V_1 and V_2 estimates for five published agentic benchmark configurations, using values derived from the cited literature. V_3 and V_{total} are not reported: three benchmarks use a deterministic single-rule grader for which reliability is not a defined quantity (no second rater), and the fourth (WebArena original) has no published grader-vs-human measurement for its partial LLM-judgment component. $V_2^{\dagger}$: real environment, no LLM user simulator.

Benchmark	V_1	V_2	V_3
τ-bench retail (SAE users)	0.62	0.88	not applicable (deterministic grader)
τ-bench retail (AAVE users)	0.62	0.80	not applicable (deterministic grader)
OSWorld	0.72	$1.00^{\dagger}$	not applicable (deterministic grader)
SWE-bench	0.59	$1.00^{\dagger}$	not applicable (deterministic grader)
WebArena (original)	0.97	$1.00^{\dagger}$	not published (partial LLM grader)

estimates, not direct measurements. V_1 is estimated from published benchmark failure rates [15] rather than direct total variation distance measurement, because the ideal construct distribution $D_{\mathcal{C}^*}$ is a theoretical object not directly observable. V_2 is operationalized via Expected Calibration Error rather than $\text{ICC}(A, 1)$ (specifically, $V_2 \approx 1 - \text{ECE}_{\text{Human-LLM}}/100$ from Seshadri et al. 16); this approximation holds only when ECE measures uniform miscalibration, which is not guaranteed. Appendix B (§B.5) works a fully hypothetical numerical example of the compounding formula for illustration; it is not tied to any specific published benchmark.

Two findings emerge. First, V_1 and V_2 are independently measurable and independently show real degradation: task validity flaws in the majority of surveyed benchmarks (Section 4) and up to 9 percentage points of simulator miscalibration with a demographic gradient (Section 5). These layers are not the missing piece; they are measured, published, and ignored by the benchmarks that report scores derived from them.

Second, and more consequentially: of the five benchmark configurations surveyed here, none has a construct-appropriate, benchmark-specific measurement of judgment-layer validity. Three have no defined V_3 at all under any IRR-based model, because their graders are deterministic single rules with no second rater. The fourth has a published grader-vs-human estimate available only by analogy from an unrelated benchmark, and only for a minority of its tasks. This is a stronger and more specific claim than "V_3 is usually low": for the large majority of widely-used agentic benchmarks, the question the compounding model poses at the judgment layer is not merely unanswered,

it is not currently answerable with a reliability metric at all, and a different validity question (does the deterministic rule correctly operationalize the construct?) applies instead, one this model does not yet formalize.

7.3. Implications for Published Benchmark Claims

When a paper reports that "Agent X achieves 82% on [benchmark]," the epistemic weight of that claim is conditioned on the validity of the three pipeline layers. Table 3 shows that for five widely-used agentic benchmark configurations, the judgment layer's validity has never been assessed in a form this model can quantify: three use a deterministic grader for which reliability is not a defined question, and the fourth's partial LLM-judgment component has no published grader-vs-human measurement. An 82% score from any of these benchmarks does not tell us that the agent achieves 82% of the intended capability, because the conversion from raw score to capability estimate requires an assessed V_3 that does not currently exist for any of them.

This does not mean the score is uninformative, nor does it license a bound such as $V_{\text{total}} \leq 0.05$: the deterministic benchmarks' V_3 is not a well-defined quantity to substitute into the compounding formula at all, rather than a known-low one, and asserting a numeric bound here would repeat the same error this paper identifies elsewhere. The correct response to an unassessed judgment layer is to say so, not to assume validity by default and not to manufacture a number where none currently exists. A benchmark whose scores carry an assessed judgment-layer validity would require, for benchmarks like τ-bench, OSWorld, and SWE-bench, the construct-validity question this model does not yet formalize (does the deterministic rule correctly operationalize the target construct?), and, for WebArena, a grader-vs-human measurement specific to its own `fuzzy_match` implementation rather than one borrowed from an unrelated benchmark.

8. Prescriptions for Valid Agentic Evaluation

Items supported by existing literature are stated as requirements; items that extend beyond current literature are marked *we propose*.

1. **Validate simulation calibration before trusting evaluation results.** Use a held-out set of real human interactions to measure the $\text{ICC}(A, 1)$ between simulated and real agent outcomes. If calibration is not validated, the evaluation signal is of unknown reliability regardless

of how precisely the judging rubric is specified [16]. Report the calibration ICC alongside any benchmark result that uses user simulation. *We propose* $ICC(A,1) \geq 0.70$ *as a minimum threshold for simulation calibration. The value is a round-number convention set just above Krippendorff [7]'s general content-analysis minimum of 0.67, not a quantity derived from the simulator literature; what the evidence supports is the direction, that LLM simulators exhibit systematic behavioral artifacts absent from trained-human rater contexts [16] and so should not be held to a lower bar than human coders, rather than the specific 3pp increment.*

2. **Select the IRR metric based on pipeline structure, not convention.** Apply Cohen's κ only when exactly two fixed raters score every item. Apply Fleiss' κ when a fixed-size panel of three or more raters scores nominal or categorical outcomes on a complete matrix. Apply Krippendorff's α when the rubric is ordinal or continuous, when missing ratings are possible, or when rater panel membership varies. Using Cohen's κ outside its structural assumptions is a silent validity failure (Section 6).
3. **Report both Fleiss' κ and Krippendorff's α for any rubric using ordinal or interval scales.** When the two metrics diverge, the divergence is itself diagnostic: $\kappa < \alpha$ suggests raters are frequently one scale point apart rather than at opposite ends of the range, a much healthier disagreement pattern than the κ score alone would imply.
4. **Use cross-family judge ensembles.** Self-preference and family-level bias in LLM judges [36, 37] are large enough to distort comparative agent rankings. Evaluation pipelines should use judges from at least two distinct provider families and report inter-judge agreement across families. Single-family judge panels produce spuriously inflated within-family agreement.
5. **Apply domain-appropriate reliability thresholds.** For exploratory capability evaluation, $\alpha \geq 0.67$ is the minimum consistent with Krippendorff [7, pp. 241–243]'s own recommendation. For safety, bias, fairness, and risk scoring, *we propose* $\alpha \geq 0.70$. For scoring that gates model deployment, compliance certification, or regulatory reporting, *we propose* $\alpha \geq 0.80$. When these thresholds cannot be met, as Jafari et al. [38] (preprint, under review at time of submission) demonstrate for LLM mental health response evaluation (three certified psychiatrists produced ICC = 0.087–0.295 and $\alpha = -0.203$), the correct response

is not to lower the threshold but to recognize that the construct is insufficiently specified for reliable measurement.

6. **Validate dynamically generated task distributions against curated reference sets.** When tasks are generated by an LLM, validate that the generated distribution samples the intended construct by testing a sample of generated tasks against domain-expert judgment: present tasks to experts who rate each on whether it faithfully instantiates the target construct, and compute V_1 as the proportion rated as construct-valid [15]. Where a curated expert reference set exists, total variation distance between the generated and reference distributions provides a more formal measure; in practice, expert-rated proportions are the tractable operationalization (Section 9, Step 1).
7. **Stratify evaluation by demographic and linguistic group.** Simulation calibration failures are not uniformly distributed across user populations [16], and LLM judge calibration failures show the same pattern across dialect groups [35]. AAVE speakers, Indian English speakers, and other non-SAE populations must be included in calibration validation before benchmark results are considered representative.
8. **Report IRR as a required field in evaluation documentation.** Standardized reporting frameworks such as EvalCards [10] provide the infrastructure to enforce consistent disclosure across papers and model releases. Including IRR metric, threshold, and metric-selection rationale as required fields in evaluation reporting cards would institutionalize the prescriptions above at the submission level.

9. Applying the Framework: A Pipeline Assessment Procedure

The following procedure operationalizes the compounding validity model as a knowledge-based pipeline assessment tool. Pipeline operators can apply it at evaluation design time to estimate V_{total} and identify which layer requires the most attention before results are reported.

1. **Estimate V_1 (task generation validity).** Generate a sample of 100 tasks from the pipeline's task generator G. Present them to domain experts who score each task on whether it faithfully samples the intended construct. Compute V_1 as the proportion rated as construct-valid. If a curated reference set is available, measure total variation distance between generated and reference distributions instead.

2. **Estimate V_2 (simulation calibration).** Run a matched subset of tasks (minimum 50) with both the LLM simulator S and real human users. Compute ICC$(A, 1)$ between simulated and real agent success rates, separately for at least two user population groups (e.g., SAE and one non-SAE group). Use the population-weighted average as V_2.
3. **Estimate V_3 (judgment validity).** Apply the IRR metric selection procedure (Figure 1) to identify the structurally correct metric for the pipeline's rater design. Compute that metric on a sample of ratings against an independent human reference. This value is V_3. If the judgment layer is a single deterministic rule (database-state comparison, unit-test execution, substring matching), V_3 as an IRR quantity is undefined (Section 7.2); a grader-vs-human validation study of the kind reported by Gurram [27] must be conducted for that grader before Step 4 can be completed.
4. **Compute the bound.** $V_{\text{total}} \leq V_1 \times V_2 \times V_3$. Report this bound alongside any benchmark result derived from the pipeline.

Interpretation guideline. If $V_{\text{total}} \leq 0.50$, benchmark scores should not be used for consequential deployment decisions without independent validation. If $V_{\text{total}} \leq 0.30$, the pipeline should be redesigned before any deployment claim is made. The formal definitions and the same-provider estimator-bias discussion are in Appendix B.

10. Threats to Validity

We assess threats to the validity of this study following the construct, internal, external, and conclusion validity framework standard in empirical research methodology [39, 40].

10.1. Construct Validity

Operationalization of V_1, V_2, V_3. The approximations underlying each V_i estimate and their implications for Table 3 are stated in the "Caveats on derived estimates" note directly beneath that table (Section 7.2). In summary: V_1 is estimated from published benchmark failure rates rather than direct total variation distance; V_2 uses Expected Calibration Error as a proxy for ICC$(A, 1)$, an approximation valid only under uniform miscalibration; and V_3 is not reported for any benchmark in that table, because it is not a well-defined quantity for a deterministic single-rule grader and has not been published for

the one partial LLM grader (Section 7.2). Any V_{total} computed from inputs of this kind should be read as a derived bound under these stated assumptions.

IRR coding taxonomy. The four-dimension coding scheme (Section 6.2) may not capture all relevant failure modes. In particular, papers that apply the structurally correct metric but with an incorrect threshold, or that report IRR on a subset of items not representative of the full evaluation, fall outside the coding scheme and may be classified as "correct" when they should not be. Appendix A provides the per-paper coding table (Table A.2) to support independent assessment.

Compounding model. The multiplicative validity bound (Equation 1) is presented as a conceptual model, explicitly not a proved theorem (Appendix B, §B.3). The motivating argument assumes layer independence and treats each V_i as a normalized signal-to-noise ratio; both assumptions are approximations. The bound should be read as a formalization of the compounding intuition, not a mathematical guarantee.

10.2. Internal Validity

Coding reliability. The literature scan (Section 6.2) was coded by the primary author. To address the potential self-referential tension (a paper about IRR using single-rater coding), we conducted a four-rater reliability study over the whole corpus. Three LLMs from distinct model families (NVIDIA Nemotron-Ultra-550B, Google Gemma-4-31B, Alibaba Qwen3-80B-A3B; OpenRouter 18) independently applied the coding instrument to the complete extracted text of each of the 55 papers, five times each, with no access to the primary-rater codes. Agreement is high on the metric-identification dimension (Q1; Krippendorff's $\alpha = 0.834$; Fleiss' $\kappa = 0.818$–0.846) and materially lower on the structural-validity judgment (Q4) that produces this paper's headline figure ($\alpha = 0.536$, or 0.549 for the binary correct-versus-misuse collapse), which falls below the paper's own exploratory threshold ($\alpha \geq 0.67$). We therefore treat the 80% IRR misuse rate as a structured expert estimate carrying a measured reproducibility range of 60.7%–84.4%, rather than as an IRR-validated point estimate. What the study validates is narrower than the headline claim: independent raters reliably recover *which* coefficient a paper computed, and agree much less on *whether* that choice constitutes misuse. This is why Section 6.2 leads with the count of papers reporting no chance-corrected coefficient at all, 26 of 55: that quantity rests on the dimension the study does validate, and the argument of this paper does not need the contested one.

The same study supplies a second caution about its own raters. On the released grounding probe, 8–10% of the quotations the raters offered as the evidence for a code cannot be located anywhere in the document they were coding (Section 6.2). An unlocatable quotation does not by itself make the code wrong, and the codes were not filtered on it, but it means some fraction of the agreement measured above was reached without demonstrable grounding in the stimulus. That is a limitation of this validation exercise and, at the same time, the most direct evidence in the paper for the practice it criticizes.

The disagreement is not evenly distributed across raters. On the failure-mode judgment, Gemma agrees with the author at pairwise $\alpha = 0.731$ and Nemotron at 0.604, while Qwen3-80B-A3B agrees with the author at only 0.412 and with Nemotron at 0.311. Qwen also departed most often from the instrument's own stated rule: on 28% of its no-coefficient codings (36 of 127, affecting 10 of the 26 papers it coded that way), it failed to record the corresponding absent-metric failure mode, whereas the other two raters never broke that rule. Because a rater's rule-following rate is observable only when the instrument states a rule and the codes are reported in full, this is itself an argument for Prescription 8. The divergence concentrated on papers with mixed or borderline evidence, for instance papers reporting a chance-corrected metric alongside a raw percentage figure, or applying a plausible-sounding metric to a design it does not quite fit, more than on papers reporting no metric at all. On several of these borderline papers, Qwen coded the design as structurally correct where the other three raters coded it as mismatched or incomplete. This is, at a meta level, an instance of the failure mode this paper documents: a single rater's calibration, left unreported or averaged away, would silently have shifted the apparent reliability of the underlying finding. Had we reported only Qwen's codes, or only a majority vote without the pairwise breakdown, this divergence would be invisible. We treat its visibility as a direct illustration of Prescription 8 (IRR as a required reporting field) rather than as noise to be discarded.

A secondary tension is worth naming explicitly: this paper argues that LLMs exhibit systematic miscalibration as evaluative raters (Sections 5.2 and 6), yet uses LLM raters as a validation mechanism here. These are not equivalent roles, and the result above sharpens rather than resolves the tension: one of the three LLM raters exhibited exactly the kind of miscalibration the paper warns about elsewhere, detectable here only because reliability was measured and reported in full. These results do not claim that LLMs are valid

substitutes for domain-expert human raters. The high metric-identification figure shows only that a determinate question with a printed answer can be coded reproducibly; the much lower structural-validity figure shows that the interpretive judgment this paper actually rests on cannot, and that LLM raters diverge from a human-coded baseline in ways visible only when IRR is computed and disclosed. That asymmetry is the paper's central argument applied reflexively to its own methodology. The pre-specified coding criteria, category summary, and per-paper coding table in Appendix A document the coding scheme; §A.4 lists notable individual cases referenced in the main text.

Purposive sampling and confirmation bias. The 55-paper sample was selected purposively to cover nine topic categories, not to exhaust a keyword-defined corpus. The 80% misuse rate is a finding within this purposive sample and should not be read as a field-wide prevalence estimate. A random sample from a keyword-defined corpus could produce a materially different rate: papers demonstrating IRR failures may be more visible and more frequently cited than papers with correct but underreported IRR practices, which would be underrepresented in a citation-weighted purposive selection. Additionally, the category design (including categories where correct IRR use was by design absent, such as Category A) inflates the apparent misuse rate relative to what a random corpus draw would show. A second source of inflation is the denominator: five of the 55 papers run no rating or annotation study of their own and so had nothing to report reliability for, and excluding them gives 78.0% (39 of 50), the figure we regard as the more defensible of the two (Section 6.2). The structured category definitions and transparent category membership (Appendix A) allow readers to assess whether the sampling frame is representative of their own research context, but do not transform a purposive finding into a prevalence claim.

10.3. External Validity

Sample scope. The 55 papers were drawn primarily from English-language venues (ACL, EMNLP, NeurIPS, ICLR) and arXiv preprints in cs.AI and cs.CL. Papers from other geographic regions, language communities, or non-English venues may show different IRR reporting patterns. The temporal scope (2022–2026) means that earlier work and work after the scan cutoff is not captured; IRR practices are evolving and the field may improve.

Domain of the compounding model. The three-layer model was constructed for agentic evaluation pipelines involving task generation, user simulation, and LLM judgment. Static benchmarks with fixed tasks, no simulation layer, and

human-only raters present a different threat model where only Layer 3 (IRR metric selection) directly applies. The prescriptions in Section 8 are intended to generalize across evaluation pipeline designs, but their applicability to static benchmarks should be assessed separately.

10.4. Conclusion Validity

Independence assumption. The primary conclusion of Section 7 is that validity failures compound multiplicatively. This conclusion depends on the independence assumption: that layer failures are statistically unrelated. When the same LLM provider family operates at all three layers, shared systematic biases cause the cross-pipeline estimates of V_i to overstate true validity for that specific configuration, making the independence bound optimistic for same-provider pipelines (Appendix B, §B.4). The qualitative conclusion (failures multiply rather than add) is robust to this concern because the bound holds as an upper limit regardless of provider overlap; same-provider configurations are subject to an additional source of optimism that practitioners should account for.

Derived estimates in Table 3. The V_1 and V_2 values in the benchmark table (Section 7.2) are derived from published measurements using approximations documented in the construct validity subsection above; V_3 is not reported there because it is not a well-defined quantity for a deterministic single-rule grader with no second rater (§7.2).

Table 4 below is therefore a hypothetical exercise, not an extension of Table 3's measurements: it asks how V_{total} would change *if* a reliability-style validity coefficient of a given magnitude could be assigned to the judgment layer, using τ-bench and SWE-bench's measured V_1/V_2 values as fixed inputs. It illustrates the sensitivity of the compounding bound to the judgment layer, not a claim that $\kappa = 0.049$ [27], drawn from a different benchmark's substring-matching grader, generalizes to either benchmark's actual (deterministic) grading mechanism.

The exercise shows how strongly the bound is governed by the judgment layer: with V_1 and V_2 at their measured values, any assumed $V_3 \leq 0.50$ places V_{total} below the pipeline-redesign threshold proposed in Section 9, and only a judgment-layer coefficient above roughly 0.55 (τ-bench) or 0.51 (SWE-bench) would lift the bound above 0.30. This is a statement about the arithmetic of the bound under assumed inputs, not a claim about either benchmark's actual validity, which cannot be computed until a grader-vs-human validation study exists for its specific grader (Section 7.2).

Table 4: Hypothetical sensitivity of V_{total} to an assumed judgment-layer coefficient for two representative benchmarks. V_1 and V_2 are held fixed at the values in Table 3; the three V_3 values are assumed levels (chance-level, low, moderate), not measurements of either benchmark's grader, whose V_3 is undefined under this model.

Benchmark	V_1	V_2	V_3	$V_{\text{total}} \leq$
τ-bench retail (SAE)	0.62	0.88	0.05	**0.027**
τ-bench retail (SAE)	0.62	0.88	0.30	**0.164**
τ-bench retail (SAE)	0.62	0.88	0.50	**0.273**
SWE-bench	0.59	1.00	0.05	**0.030**
SWE-bench	0.59	1.00	0.30	**0.177**
SWE-bench	0.59	1.00	0.50	**0.295**

11. Discussion

11.1. What This Paper Does Not Claim

We do not claim that current agentic evaluations are worthless. They provide directional signal. We claim that the field's current practices overstate the validity and precision of that signal, and that the standards for reporting should be substantially higher. A benchmark result reported without IRR metrics, simulation calibration evidence, and metric selection justification is incomplete, in the same way that a clinical measurement reported without instrument validity data is incomplete.

11.2. The Governance Stakes

For academic benchmark comparisons, validity limitations are a scientific quality issue. For deployment decisions, safety certifications, and regulatory claims, they are a governance failure. Agentic AI systems are increasingly being deployed in consequential contexts. The evaluation apparatus used to justify those deployments must meet the same validity standards applied to any measurement-driven governance process.

The regulatory dimension makes this concrete. Emerging frameworks such as the EU AI Act require evidence of comprehensive risk assessment for general-purpose AI models. A systematic analysis of 194,955 benchmark questions found that capabilities central to loss-of-control (including evading human oversight, self-replication, and autonomous AI development) receive zero benchmark coverage [4]. If the construct coverage gap means that no existing benchmark can provide evidence of regulatory compliance, then the

IRR validity of those benchmarks is a secondary concern: the primary problem is that they do not measure what regulations require. Both failures must be addressed together.

11.3. Broader Impact

This work raises the evidential bar for deployment certifications, surfaces demographic calibration failures that existing evaluation practice obscures, and gives practitioners eight actionable criteria they can apply using existing psychometric tools.

Three misuse risks follow from this work. First, the proposed reliability thresholds ($\alpha \geq 0.70$, $\alpha \geq 0.80$) may become compliance checkboxes rather than genuine validity indicators. Meeting a threshold does not guarantee valid measurement; it establishes the minimum condition for a score to be interpretable. Second, stratified calibration across demographic and linguistic groups increases evaluation cost; benchmark maintainers and shared evaluation platforms should bear primary responsibility for calibration coverage, not individual research teams. Third, the finding that three certified psychiatrists produced ICC = 0.087–0.295 on LLM mental health safety evaluation [38] should not discourage research in high-stakes AI domains. It means the construct is underspecified for reliable measurement, which calls for more rigorous construct development, not withdrawal.

12. Conclusion

The move from static to dynamic agentic evaluation does not reduce the importance of inter-rater reliability; it multiplies the surfaces on which reliability can fail and introduces new systematic biases at each layer: task generation, world simulation, and judgment. The compounding effect means that an evaluation pipeline with moderate validity problems at each of three layers can produce an evaluation signal that is far less valid than any single-layer estimate would suggest.

The science of measurement was not built for static benchmarks alone. It was built for situations where the measurement apparatus is a source of variance, where raters disagree in structured rather than random ways, and where the construct must be separated from the instrument measuring it. That is agentic evaluation.

These standards are achievable. El Hattami et al. [28] (WebArena Verified) demonstrates this directly: adding rigorous human annotation

with fixed annotator pairs per site and reporting Cohen's $\kappa = 0.83$ (95% CI [0.81, 0.85]) satisfies the metric selection requirement (Prescription 2), clears the deployment-gating reliability level (Prescription 5, read as the approximation noted in Section 6.5), and treats IRR as a required reporting field (Prescription 8). This is a demonstration of rigorous IRR *practice* for the construct it annotates (task-specification ambiguity and category validity), not a benchmark-specific measurement of agent-outcome judgment validity (Section 7.2); the two are separable achievements, and the field currently manages neither for most benchmarks. WebArena Verified meets three of the eight prescriptions through a design choice the field already knows how to make.

Fleiss' κ, Krippendorff's α, ICC, and construct validity frameworks are available, validated, and applicable. The field's task is to use them, to report IRR as a first-class metric alongside every benchmark result, and to recognize that a score without validity evidence is not a measurement. It is a number.

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CRediT Authorship Contribution Statement

William Caban: Conceptualization, Methodology, Formal Analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Declaration of Competing Interests

The author is employed by Red Hat, Inc. This research was conducted independently as part of the author's doctoral program at Alma Mater Europaea University. The views expressed are the author's own and do not represent the official position of Red Hat, Inc. The author declares no financial competing interests.

Declaration of Generative AI and AI-Assisted Technologies

During preparation of this work the author used Claude (Anthropic) to assist with LaTeX formatting, editorial length reduction, and improvements of the reliability-study code and analysis scripts; the author designed the study, verified all outputs, and takes full responsibility for the work. Claude is not listed as an author.

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Ethics Statement

This study is based entirely on analysis of publicly available published research. No human participants, personal data, or animal subjects were involved.

Data Availability

The data and code supporting this study are openly available at the repository below. It contains the per-paper coding table for all 55 papers in the literature scan (`full_scan_55papers.csv`, reproduced as Table A.2) with the rationale recorded for every coded paper, a machine-checkable evidence trail for the metric-reported code (`audit_report.csv`, which re-derives that code lexically from each full document and reports where in the document the evidence sits), the author codes and all five rater runs of the reliability study (`author_codes_55.csv`, `results_fulltext.csv`), and the Python scripts for extracting the stimulus, running the LLM raters, validating the returned codes, and computing Krippendorff's α (`extract.py`, `second_rater.py`, `validate.py`, `compute_alpha.py`). Because the coded papers are third-party copyrighted works, the repository publishes the SHA-256 checksum of each extracted document (`stimulus_hashes.csv`) and the pinned extractor version rather than the extracted text itself, so that a replicator who obtains the same PDFs can verify byte-identical stimulus reconstruction:

`https://github.com/williamcaban/experiment-measurement-without-validity`

The repository is released under the Apache 2.0 license. All 55 coded papers are publicly available; `manifest.csv` records the resolution scheme, versioned identifier, fetch URL, and source-PDF checksum for each, which is what makes the checksum verification above reproducible. No proprietary datasets, model weights, or benchmark systems were created or used. Code for the compounding model (Appendix B) requires no implementation beyond the equations stated.

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Appendix A. Literature Scan: Coding Criteria, Category Summary, and Per-Paper Codes

This appendix documents the methodology and results of the structured literature scan reported in Section 6.2. It presents the pre-specified coding criteria, a category summary of the 55 coded papers, the complete per-paper coding table, and notable individual cases referenced in the main text.

A.1 Pre-Specified Coding Criteria

Each of the 55 coded papers was coded on four dimensions, applied uniformly before reading the paper's results section. Two codes were revised after the initial pass, both on evidence from the mechanical lexical audit released as `audit_report.csv` rather than on reflection: one paper was found to report a coefficient for its own annotation study that the first pass had recorded as absent, and one was moved from absent to incomplete. The `notes` column of the released coding table records the revision and its date for each.

The dimensions are:

1. **IRR metric reported**: Cohen's κ, Fleiss' κ, Krippendorff's α, ICC (form specified or unspecified), percentage agreement only, or no metric reported.
2. **Rater design**: number of raters; whether rater identity was fixed across items or varied (rotating pool, crowdsourced, or multiple independent runs of the same model).
3. **Measurement scale**: binary, nominal (unordered categories), ordinal or ternary (ordered but discrete), or continuous.
4. **Structural validity**: whether the reported metric's structural assumptions were satisfied by the rater design and measurement scale. Coding applied the following rules: Cohen's κ for > 2 raters or rotating identity $\rightarrow$ MISMATCH; Fleiss' κ for ordinal or continuous scale $\rightarrow$ PARTIAL MISMATCH; percentage agreement without chance correction $\rightarrow$ INCOMPLETE; no IRR with automated grader of unvalidated accuracy $\rightarrow$ LAYER-3 VALIDITY FAILURE.

A.2 Category Summary

Table A.1 summarizes the 55 individually coded papers by topic category. The nine categories were defined before coding to cover the main structural situations in which IRR is, or should be, reported in agentic evaluation: benchmarks scored automatically (A), studies that validate such graders against humans (B), benchmarks that do report formal IRR (C), LLM-as-a-Judge studies (D), benchmarks whose reported metric does not fit their rater design (E), long-horizon and capability benchmarks (F), safety, RLHF, and preference evaluation (G), recent 2025–2026 evaluation papers (H), and safety-critical settings in which IRR failures carry direct consequences (I). Categories A, C, and E were defined by an expected IRR pattern and therefore contribute to the within-sample rate by design (Section 10.2).

Table A.1: Literature scan category summary. "Correct" = structurally valid metric with explicit rationale. "Failure mode" = dominant pattern among incorrect papers.

Cat.	Topic	Papers	Correct	Dominant failure mode
A	Major agentic benchmarks (automated grading)	9	0	No IRR; automated grader validity assumed
B	Automated grader validity studies	3	1	No IRR metric reported
C	Benchmarks with formal IRR	11	9	Krippendorff's α applied to a self-consistency design; unweighted Cohen's κ on an ordinal scale
D	LLM-as-a-Judge studies	4	0	Cohen's κ aggregated over a 54-model panel; IRR metrics applied to self-consistency designs
E	Benchmarks with structural metric mismatch	5	0	Fleiss' κ for ordinal rubrics; % only
F	Long-horizon and capability benchmarks	5	0	No metric reported; % agreement only
G	Safety, RLHF, and preference evaluation	8	0	No IRR metric (ELO, Pearson, or none); % agreement only
H	Recent 2025–2026 evaluation papers	3	0	No metric; weighted Cohen's κ still not the prescribed metric
I	Safety-critical IRR failures	7	1*	Raw % agreement; coefficient reported without a stated threshold
Total		**55**	**11 (20%)**	

* Category I includes one paper (Jafari et al. 38) that uses the correct metrics but reports catastrophically low values (ICC = 0.087–0.295, $\alpha = -0.203$).

A.3 Per-Paper Coding Table

Table A.2 lists every paper in the scan with its four codes. The year shown is the year of the version that was coded, which `manifest.csv` pins to an exact identifier and checksum; where a work was subsequently published, the reference list cites that version of record instead, so the two can differ by a year for the same paper. Q1 (metric reported): KCO = Cohen's κ; KFL = Fleiss' κ; KAL = Krippendorff's α; ICC = intraclass correlation; PA = percent agreement only; NIL = no recognized IRR metric. Q2 (rater design): F2 = two fixed raters, same pair on every item; FN = fixed panel of three or more; RV = rotating or varying rater identity; CS = crowdsourced or open voting pool; UNK = not described or no rater design. Q3 (scale): BIN, NOM, ORD, CON, UNK. Q4 (structural validity): OK = metric fits design and scale with threshold and rationale; MM = structural mismatch; INC = percent agreement only, or correct family without threshold or rationale; ABS = no IRR metric reported. The rationale for each coded paper is recorded in the `notes` column of the released coding table, and `audit_report.csv` independently re-derives the metric-reported code from each full document so that code can be checked mechanically against its source.

Table A.2: Per-paper coding of the 55-paper literature scan. Category letters follow Table A.1.

#	Paper	Cat.	Q1	Q2	Q3	Q4
1	Yao et al. (2024), τ-bench [21]	A	NIL	UNK	BIN	ABS
2	Xie et al. (2024), OSWorld [22]	A	NIL	UNK	BIN	ABS
3	Gurram (2026), AgentProp-Bench [27]	B	KCO	F2	BIN	OK
4	El Hattami et al. (2025), WebArena Verified [28]	C	KCO	F2	BIN	OK
5	Megahed et al. (2025), Reliable decision support with LLMs	C	KAL	FN	BIN	OK
6	Yamauchi et al. (2025), Empirical study of LLM-as-a-Judge design choices	C	KAL	UNK	ORD	MM
7	Han et al. (2025), Judge's Verdict [33]	D	KCO	RV	BIN	MM
8	Jain et al. (2026), Multi-LLM thematic analysis with dual reliability metrics	D	KCO	UNK	NOM	MM
9	Fan et al. (2026), AgentProcessBench [31]	E	KCO	F2	ORD	MM
10	Akkil et al. (2026), Emergence WebVoyager	E	PA	F2	BIN	INC
11	Hu, T. et al. (2025), Multi-agent debate for LLM judges	F	NIL	FN	CON	ABS
12	Zhou, H. et al. (2026), Taxonomic bias evaluation of LLM judges	F	NIL	RV	ORD	ABS
13	Movva et al. (2024), Annotation alignment (conversational safety) [29]	G	NIL	CS	ORD	ABS
14	Chiang et al. (2024), Chatbot Arena [30]	G	NIL	CS	BIN	ABS
15	Chandwani et al. (2026), LH-Bench	H	KCO	FN	ORD	MM
16	Ouyang et al. (2022), InstructGPT [34]	I	PA	CS	ORD	INC
17	Jafari et al. (2026), Expert evaluation in mental-health AI safety testing [38]	I	ICC	FN	ORD	OK

continued on next page

Table A.2 (continued)

#	Paper	Cat.	Q1	Q2	Q3	Q4
18	Piot et al. (2025), LLM reliability in hate-speech detection (xRR)	I	KCO	RV	BIN	MM
19	Kunilovskaya et al. (2026), Who annotates in NLP?	I	KAL	RV	NOM	INC
20	James (2026), Counting on consensus (IAA metric selection) [14]	I	NIL	UNK	UNK	ABS
21	Jimenez et al. (2024), SWE-bench [23]	A	NIL	UNK	BIN	ABS
22	Yang et al. (2026), HarmMetric Eval	B	NIL	UNK	BIN	ABS
23	Ganguli et al. (2022), Red teaming language models to reduce harms [32]	E	KFL	FN	ORD	MM
24	Seshadri et al. (2026), Lost in simulation [16]	F	NIL	UNK	UNK	ABS
25	Meimandi et al. (2025), Measurement imbalance in agentic AI evaluation [3]	G	PA	F2	BIN	INC
26	Zhu et al. (2025), Best practices for rigorous agentic benchmarks [15]	H	NIL	UNK	UNK	ABS
27	Mialon et al. (2023), GAIA [24]	A	NIL	UNK	UNK	ABS
28	Zhou, S. et al. (2023), WebArena [25]	A	NIL	UNK	UNK	ABS
29	Barres et al. (2025), τ^2-Bench [26]	A	NIL	F2	UNK	ABS
30	Faisal et al. (2024), Dialectal toxicity detection [35]	I	PA	UNK	ORD	INC
31	Spiliopoulou et al. (2025), Play Favorites (self-bias in LLM-as-a-Judge) [36]	C	KAL	FN	ORD	OK
32	Wataoka et al. (2024), Self-preference bias in LLM-as-a-Judge [37]	D	NIL	UNK	UNK	ABS
33	Bhardwaj et al. (2024), State of data curation at NeurIPS	C	ICC	RV	ORD	OK
34	Pradhan et al. (2025), LLM-as-a-Judge for legal document recommendation	C	KAL	F2	ORD	OK
35	Marino et al. (2025), AIReg-Bench	C	KAL	RV	ORD	OK
36	Guo et al. (2025), MCP-AgentBench	C	KFL	FN	BIN	OK
37	Mazaheri et al. (2026), AgentAtlas	I	NIL	UNK	UNK	ABS
38	Borse et al. (2025), IRR between LLMs and human raters in qualitative analysis	C	KCO	F2	BIN	OK
39	Xue et al. (2025), An illusion of progress? (web agents)	F	PA	F2	BIN	INC
40	Liu, P. et al. (2026), RLHF: a statistical perspective	G	NIL	UNK	UNK	ABS
41	Zhao, W. et al. (2024), Challenges in trustworthy human evaluation of chatbots	E	KFL	FN	ORD	MM
42	Coyne (2026), Three models of RLHF annotation	G	NIL	UNK	UNK	ABS
43	Hu, Z. et al. (2024), Explaining length bias in LLM-based preference evaluations	G	KFL	RV	BIN	INC
44	Frick et al. (2024), How to evaluate reward models for RLHF	G	NIL	CS	BIN	ABS
45	Liu, S. et al. (2025), How humans help LLMs (preference annotators)	G	NIL	UNK	UNK	ABS
46	Sharma et al. (2025), Consistent detection of cognitive distortions	E	KFL	RV	NOM	MM
47	Keith (2025), LLM-as-a-Judge proxies for mathematical coherence	D	ICC	UNK	CON	MM
48	Wu et al. (2026), Agent-based software artifact evaluation	C	KCO	F2	ORD	MM
49	Kwa et al. (2026), Measuring AI ability to complete long software tasks (METR)	A	NIL	UNK	UNK	ABS
50	Lee et al. (2026), Meta-Harness	F	NIL	UNK	UNK	ABS
51	Rabinovich et al. (2026), Near-Miss (latent policy failure detection)	B	NIL	UNK	BIN	ABS
52	Yehudai et al. (2026), Survey on evaluation of LLM-based agents	H	NIL	UNK	UNK	ABS

continued on next page

Table A.2 (continued)

#	Paper	Cat.	Q1	Q2	Q3	Q4
53	Libon et al. (2026), ResearchArena	A	NIL	UNK	UNK	ABS
54	Sun et al. (2026), AgenticDataBench	A	NIL	UNK	UNK	ABS
55	Wiyono et al. (2025), IndoPref	C	KAL	F2	BIN	OK

A.4 Notable Individual Cases

- Jafari et al. [38]: Three certified psychiatrists, correct metrics, ICC = 0.087–0.295 and $\alpha = -0.203$ on LLM mental health response safety: highest-stakes domain, most disagreement.

- Ouyang et al. [34]: 40 rotating contractors, 4-way preference ranking. Reports 72–77% raw percent agreement with no chance correction; requires Krippendorff's α for rotating-rater ordinal design.

- Gurram [27]: Direct grader-vs-human comparison. Substring grading achieves $\kappa = 0.049$ (chance level), a 3-LLM ensemble $\kappa = 0.432$, a single GPT-4o-mini judge $\kappa = 0.567$, against $\kappa = 0.835$ between the two human annotators.

- El Hattami et al. [28] (WebArena Verified): One primary annotator per site with a paired blind verifier (four annotators across the four self-hosted sites; Map is not self-hosted and is excluded from their reliability analysis), task-specification-ambiguity and category-validity labels, Cohen's $\kappa = 0.83$ (95% CI [0.81, 0.85]) task-weighted across all 812 tasks: correct metric, threshold exceeded, reporting complete. The strongest positive example of IRR *practice* in the scan, though the labeled construct is benchmark-audit quality, not agent-outcome judgment (Section 7.2).

A.5 Production Rules for IRR Metric Selection

Table A.3 states the decision procedure of Figure 1 as a rule base. It is given here rather than beside the figure so that the body carries the procedure and the appendix carries its formal statement; the two are the same artifact, and a practitioner needs only the figure.

Table A.3: Production rules corresponding to the terminal nodes of the IRR metric selection decision tree (Figure 1). Rules are applied in order; the first matching rule determines the metric. R3 and R4 apply whether the panel is fixed or rotates, provided the number of raters per item is constant (§2.2); R2 is reserved for designs where that count itself is irregular.

Rule	Condition	Metric
R1	IF rater_count = 2 AND identity_fixed = true AND complete_matrix = true AND scale = nominal/binary	Cohen's κ
R1†	IF rater_count = 2 AND identity_fixed = true AND complete_matrix = true AND scale = ordinal	Weighted κ or Krippendorff's α (ordinal)
R1‡	IF rater_count = 2 AND identity_fixed = true AND complete_matrix = true AND scale $\in$ {interval, continuous}	ICC$(A, 1)$
R2	IF missing_ratings = true OR raters_per_item_varies = true	Krippendorff's α (general)
R3	IF rater_count $\geq$ 3 AND complete_matrix = true AND raters_per_item_constant = true AND scale $\in$ {nominal, binary}	Fleiss' κ
R4	IF rater_count $\geq$ 3 AND complete_matrix = true AND raters_per_item_constant = true AND scale $\in$ {ordinal, interval, continuous}	Krippendorff's α (ordinal/interval distance)

Appendix B. Formal Derivation of the Compounding Validity Model

This appendix provides the mathematical treatment supporting Section 7.

B.1 Definitions

Let an agentic evaluation pipeline Π consist of three sequential components:

- G: a task generation function that samples tasks t from a distribution D_G over task space $\mathcal{T}$
- S: a simulation function that, given a task t and an agent A, generates an interaction trajectory $\tau = S(t, A)$ drawn from a distribution $D_S(t)$
- J: a judgment function that, given a trajectory τ, assigns a score $s = J(\tau) \in \mathbb{R}$

Let $\mathcal{C}^*$ denote the target construct: the latent capability the evaluation is designed to measure. The construct $\mathcal{C}^*$ induces an ideal task distribution $D_{\mathcal{C}^*}$ over $\mathcal{T}$ and an ideal scoring function J^* that maps agent behavior to true capability.

B.2 Layer-Specific Validity Measures

V_1: Task Generation Validity..

$$V_1 = 1 - d_{\mathrm{TV}}(D_G,\, D_{\mathcal{C}^*})$$

where d_{TV} denotes total variation distance. $V_1 = 1$ when $D_G = D_{\mathcal{C}^*}$ exactly; $V_1 = 0$ when the supports are disjoint.

V_2: Simulation Calibration Validity..

$$V_2 = \mathrm{ICC}(\mathrm{outcomes}_{\mathrm{sim}},\ \mathrm{outcomes}_{\mathrm{real}})$$

where ICC is computed using the two-way mixed-effects model for absolute agreement ($\mathrm{ICC}(A, 1)$). V_2 must be estimated separately per demographic and linguistic user group:

$$V_2 = \mathbb{E}_{g \sim P_{\mathrm{users}}}[V_2^{(g)}]$$

V_3: Judgment Validity.. V_3 measures the degree to which the rating protocol produces valid inter-rater agreement, using the appropriate IRR metric for the pipeline's measurement scale.

B.3 The Multiplicative Validity Bound

Conceptual bound. We state the following as a conceptual model, not a proved mathematical proposition. Under the assumption that V_1, V_2, V_3 are independently determined:

$$V_{\text{total}}(\Pi) \leq V_1 \cdot V_2 \cdot V_3$$

Motivating argument. Let q denote an agent query drawn from the evaluation pipeline. Under independence, the signal-to-noise ratio of the pipeline degrades multiplicatively across layers:

$$\text{SNR}(\Pi) \leq \text{SNR}(G) \cdot \text{SNR}(S) \cdot \text{SNR}(J)$$

Treating each V_i as the normalized SNR of the corresponding component then yields the bound. The motivating argument is that layer-wise validity losses multiply; this intuition is well-grounded even if the full formal derivation is not supplied here.

Remark 1. *Because this is a conceptual bound rather than a proved proposition, equality in the inequality is not established. The bound is tightest (most informative) when the cross-pipeline V_i estimates accurately reflect the specific pipeline being assessed. When shared systematic biases are present (for example, when the same provider family operates at all three layers), the cross-pipeline estimates are optimistic, making the bound itself loose (see §B.4).*

B.4 Estimator Bias in Same-Provider Pipelines

The point estimates V_1, V_2, V_3 used in practice (e.g., Table 3) are drawn from studies conducted on pipelines that differ from the one being assessed. These estimates are therefore cross-pipeline measurements, not direct measurements of the specific pipeline under evaluation.

When the same provider family operates at all three layers of a specific pipeline, systematic tendencies (verbosity preference, formatting bias, positional sensitivity) manifest simultaneously at G, S, and J. Because the cross-pipeline estimates were obtained on pipelines without this shared bias, they do not capture the additional validity loss attributable to within-family amplification. The result is that the external V_i estimates overstate the true layer validity for that specific configuration, making the bound

$V_{\text{total}} \leq V_1 \cdot V_2 \cdot V_3$ optimistic: the true V_{total} for a same-provider pipeline may fall below the bound implied by the independently-sourced estimates.

This effect operates through *estimator bias*, not through statistical dependence among the V_i as random variables. The bound in Equation (1) remains valid as an upper bound regardless of provider overlap; the point is that same-provider configurations produce cross-pipeline estimates that are systematically too high, so the bound, while mathematically correct, is a loose one.

Mitigation. Using distinct provider families for task generation (G), simulation (S), and judgment (J) ensures that shared systematic biases are not amplified across layers. Cross-provider pipeline designs also allow cross-family agreement to be measured directly as a validity check on the V_3 estimate.

B.5 Worked Numerical Example

The values below are illustrative, not measurements. V_1 and V_2 are plausible figures of the kind reported in Table 3, while the two V_3 values are *assumed* levels: no benchmark in this survey publishes a judgment-layer coefficient, and for a deterministic grader V_3 is undefined under this model (Section 7.2). The example shows how the bound propagates under an assumption about V_3, not what any pipeline's validity is.

Cross-provider design: $V_{\text{total}} \leq 0.70 \times 0.75 \times 0.75 = 0.394$

Within-family design with metric mismatch: $V_{\text{total}} \leq 0.70 \times 0.75 \times 0.58 = 0.305$

An evaluation score reported to two decimal places from a pipeline in the 0.30–0.40 validity range is precise to a degree of resolution the underlying measurement cannot support.

B.6 Estimation Protocol

The practical estimation protocol for V_1, V_2, V_3, and V_{total} is presented in full as Section 9 in the main body. The steps in that section implement the formal definitions in B.1–B.2 using tractable approximations: expert-rated sampling for V_1, ICC$(A, 1)$ between simulated and real user outcomes for V_2 (estimated separately per demographic group), and the structurally correct IRR metric from Figure 1 for V_3. The interpretation guideline ($V_{\text{total}} \leq 0.50$: no consequential decisions without independent validation; $V_{\text{total}} \leq 0.30$: pipeline redesign required) accompanies the protocol in Section 9.